

Empathy Map Canvas

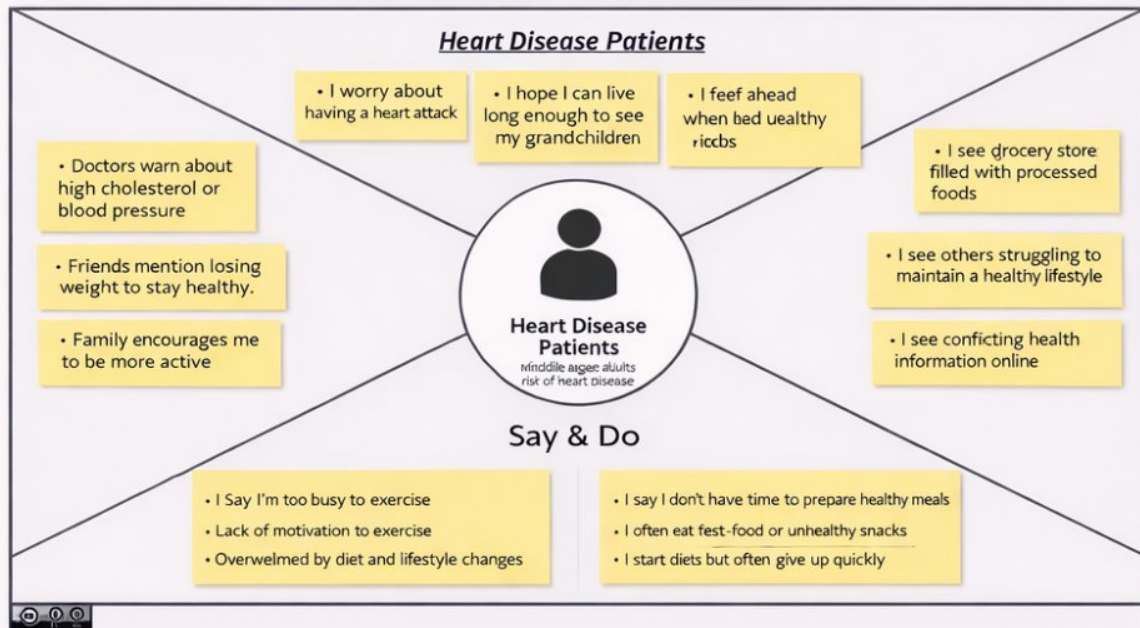
Heart Disease Analysis

An empathy map canvas is a simple, easy-to-digest visual that captures knowledge about a user's behaviors, challenges, and attitudes.

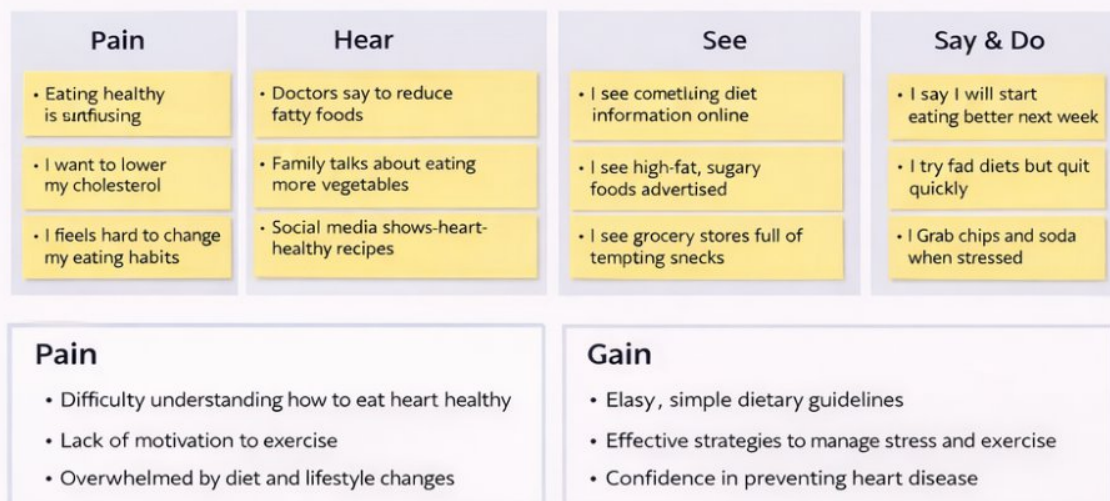
It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Heart Disease Analysis Empathy Map



Example: Diet & Heart Health Application



Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	
Project Name	
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per table1 & table 2

Example: Heart Disease Monitoring Application

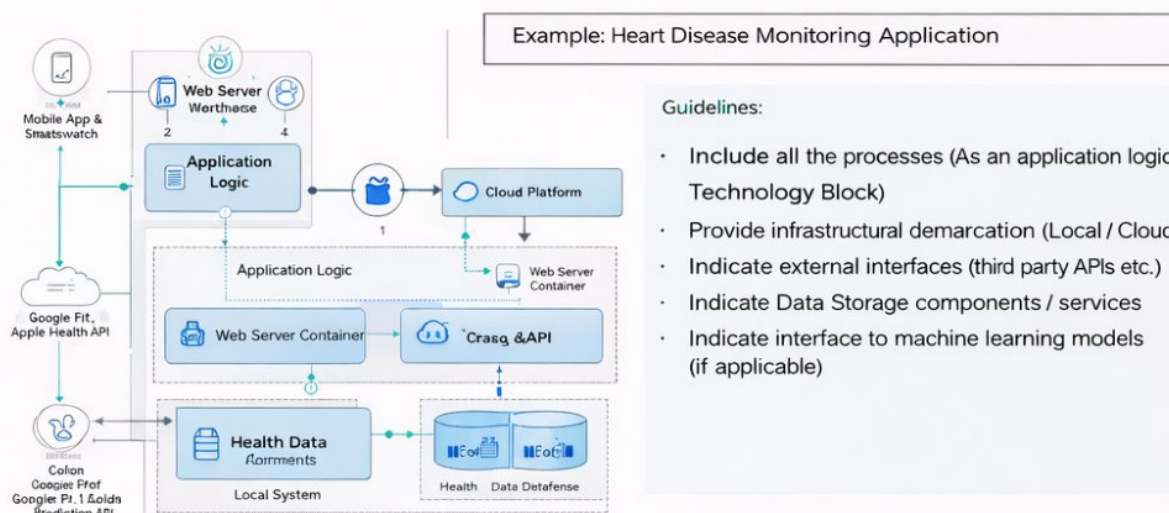


Table 1: Technology Stack

Table 1: Technical Standard requirements of the proposed Heart Disease Monitoring System.

S.No.	Component	Description	Technology
1.	User Interface	User Interface; asse asses the use; in delemaps.or: Flutter	HTML, CSS, JavaSct / Flutter / reactnative / T / Machine / neurhaacy
2.	Application Logic	Provide users to connect matication-report-elavibgy/ustrg	Sython; Prana/ rela, Sython / [otil Pune]
3.	External API-1	Provide external interfaces:: Google Fit API, Apple Hsalndila	Python Tenterat Bestesle Ellutter
4.	Database	Provide external interfaces:: CDC Blood Pressure / Nutrition API	CDC Blood Pressure / Nutrition API
5.	External API-2	Provide external interfaces: Heart Risk Prediction Model	Health Data Database / Pseudocryptis / [reicobay
6.	Infrastructure	Application Deployment, escafi te A/AWS; Docker, Kubermestes	AWS, Docker, Kubernetes

Table 2: Application Characteristics:

S.No:	Characteristics	Description	Technology
1.	Open-Source Frameworks	List open-source frameworks used	Flutter, Tensorflow.
2.	Security Implementations	List all security, / access controls (e.g. of 2FA, login).	AES-256, Encryptions, IAM Authentication, Atlassian
3.	Scalable Architecture	Justify runtime scalability - (3-tier, micro-services).	2-tier, micro-services, etc.
4.	Availability	Justify the availability of distributed servers, etc.	Technology used

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	
Project Name	
Maximum Marks	4 Marks

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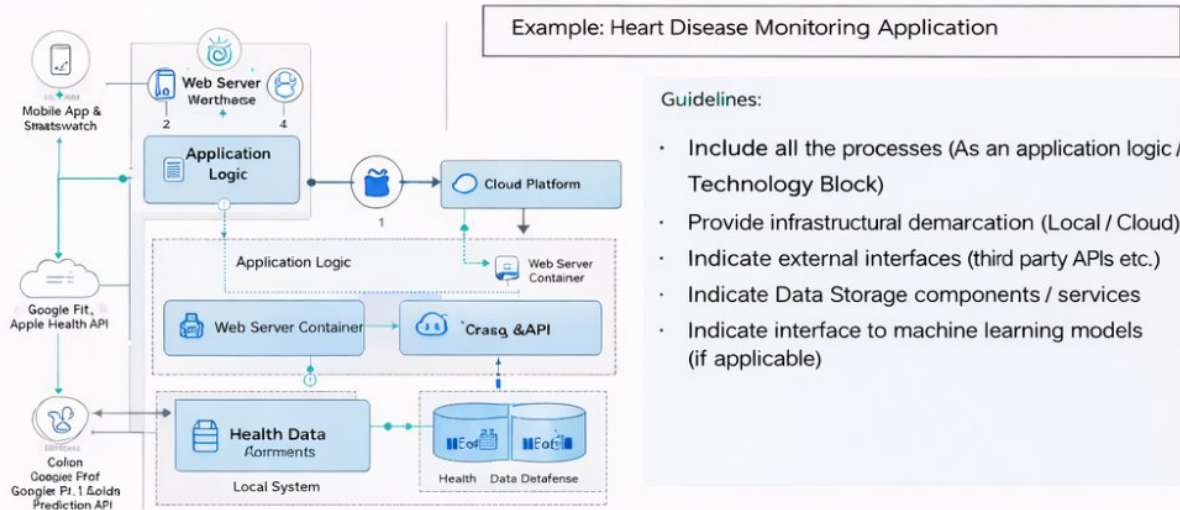


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Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	
Project Name	
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed Heart Disease Monitoring System.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	- Registration through Form - Registration through Gmail - Registration through Apple Health - Registration through Google Fit
FR-2	User Confirmation	- Confirmation via Email - Confirmation via OTP
FR-3	Health Data Logging	- Log Age, Blood Pressure, Cholesterol, etc. - Sync activity data from fitness tracker
FR-4	Heart Risk Prediction	- Analyze user data to generate heart risk score - Provide personalized recommendations

Non-functional Requirements:

Following are the non-functional requirements of the proposed Heart Disease Monitoring System.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	User-friendly, intuitive interface.
NFR-2	Security	Data privacy, HIPAA compliance.
NFR-3	Reliability	Accurate, consistent heart risk analysis.
NFR-4	Performance	Fast processing of health data inputs.
NFR-5	Availability	High uptime and responsive support.
NFR-6	Scalability	Able to handle growing user base.

Customer Problem Statement Template:

Create a problem statement to understand your customer’s point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you’ll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe the customer and their attributes here
I’m trying to	List the thing they are trying to achieve here
but	Describe the problems or barriers that get in the way here
because	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions the result from experiencing the problems or barriers

Example:

I am	I’m trying to	But	Because	Which makes me feel
a traveler	book flights on my phone	it takes a long time	The current airline website are slow and not optimized for mobile access.	Frustrated

Problem Statement (PS)		I am (Customer)	I’m trying to	But	Because	Which makes me feel
PS-1	a traveler	a traveler	book flights on my phone	it takes long time	The current airline websites are slow and	Frustrated
PS-2	a busy mom	a busy mom	quickly order groceries online for my family	It's confusing no. vingaite	not optimized for mobile access. It's unclear how to sedterme to cart and hack out	

Heart Disease Analysis Template:

Heart disease analysis focuses on studying various factors that contribute to cardiovascular diseases such as high blood pressure, cholesterol levels, lifestyle habits, and genetic risks. The goal of this analysis is to identify patterns, risk factors, and preventive strategies that can help reduce the occurrence of heart diseases.

This project encourages collaborative research and analytical thinking to evaluate medical data and identify possible causes and solutions for heart-related conditions. Through systematic analysis and prioritization of risk factors, teams can better understand how lifestyle, diet, stress, and medical conditions influence heart health.

Step-1: Team Gathering, Collaboration and Select the Research Problem



Heart Disease Analysis

Use this template to conduct research questions and then identify the teams in team to planning, analyzing, and participating can team efforts, give to you're sitting in the same room.

- ⌚ 10 minutes to prepare
- 📅 1 hour for sessions
- 👥 2-8 people recommended

4

Before you collaborate

A small amount of preparation helps you your team conduct a productive research session.

⌚ 16 minutes

Team gathering

Defining members who will participate research the research assign roles same data analyst, researcher, and presenter.

Set the research goal

- Identifying major risk factors of heart disease
- Analyzing patient data patterns
- Understanding prevention methods

Learn the research tools

Use statistical analysis tools, medical datasets, and research articles to support your study.

8

Define your problem statement

- How do lifestyle factors influence heart disease risk?
- What is the relationship between cholesterol levels and heart disease?
- How does age and blood pressure affect the probability of heart disease?

Key rules of effective research brainstorming

1. Stay focused on heart disease analysis.
2. Encourage data-based ideas and insights.
3. Listen to each team member's research findings.
4. Avoid judging ideas during the brainstorming stage.
5. Collect as much relevant information as possible.
6. Use charts, graphs, or datasets to visualize results.



Key rules of effective research brainstorming

- ① Stay focused on heart disease analysis.
- ② Encourage data-based ideas and insights.

Step-2: Brainstorm, Idea Listing and Grouping



Brainstorm

- List possible ideas from your team to help address your problem statement.

⌚ 15 minutes



Group ideas

- Group ideas into key areas or themes before clustering similar or related ideas as you go. Use these groups to form categories that help you better understand the key topics and identify patterns.

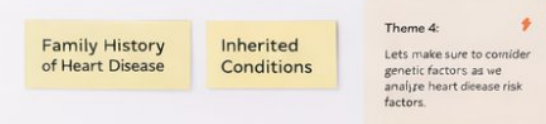
⌚ 20 minutes

Risk Factors



Patient 1	Patient 2	Patient 3	Patient 3
Cholesterol 230	Cholesterol 230	Cholesterol 220	Cholesterol 2360
BP 140/90	BP 135/85	Inactive	Age 41
Smoker	Age 37	Age 46	Age 64

Theme 4: Genetic Factors



Theme 4

Cholesterol 230	BP 140/99	Chronic	Age
BP 140/90	Smoker	Age 51-4	31+
Inactive	Age 47	Age 65	69+

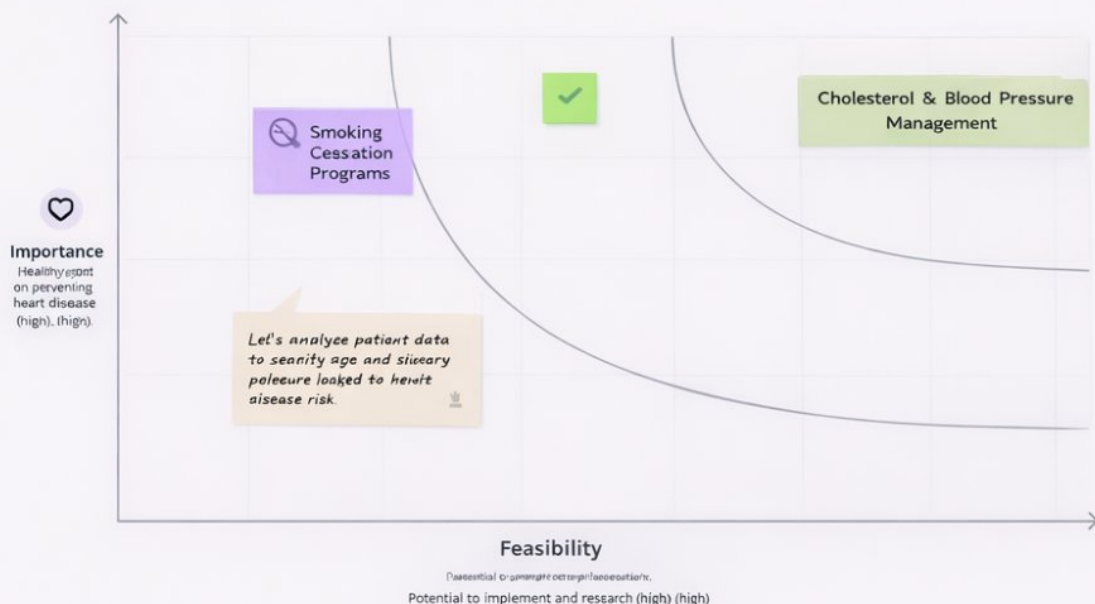
Step-3: Idea Prioritization



Prioritize

- Discuss ideas and plot on the same page after each important meeting. Visualize how impactful ideas are on a grid to determine which ideas are important and most.

⌚ 15 minutes



Heart Disease Monitoring System – Solution Fit Template

Project Title:

Team ID: PNT2022TMDxxxxxx

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - Adults aged 40+, - Working parents of 40-50, kids - Individuals with a family history heart disease	6. CUSTOMER CONSTRAINTS CC - Lack of awareness about heart disease risk - Limited time and motivation for regular health check-ups - Privacy concerns about data security	5. AVAILABLE SOLUTIONS AS - Basic fitness trackers that provide a trigger alert - Apps for diet and fitness tracking without heart disease risk prediction - Regular doctor visits, but not real-time monitoring	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&F - Monitor health metrics easily - Detect and prevent heart disease early - Receive personalized and actionable health advice	9. PROBLEM ROOT CAUSE RC - Busy lifestyles prevent regular health monitoring - Fear or ability (lack of knowledge, tools, or time) to understand heart health data - Traditional healthcare solutions are reactive, not proactive	7. BEHAVIOUR BE - Occasionally use fitness trackers to track steps/heart rate - Schedule annual doctor visits for check-ups - Look for health information online, but without "proactive monitoring"	
Identify strong TR & EM	3. TRIGGERS TR What sign/signal might trigger people into acting? Flea alert, reading about a heart disease solution in the news.	10. YOUR SOLUTION SL Our AI-driven mobile app and smartwatch system monitor health metrics like BP, cholesterol, activity levels, and predict heart disease risk. It provides personalized health advice and recommendations, fostering proactive heart health management.	8. CHANNELS of BEHAVIOUR CH Web & App Store listings Social media ads (Facebook, Instagram) Targeted online communities (Health, Wellness)	Identify strong TR & EM
	4. EMOTIONS: BEFORE / AFTER EM How will resources be emotional it wants (eg. lack of feedback addressed). Before: Nonconcerned about heart health, worried about risks. After: Confident, relieved, proactive in managing health	9. OUR SOLUTION Our AI-driven mobile app and smartwatch system monitor health metrics like BP, cholesterol, activity levels, and predict heart disease risk. It provides personalized health advice and recommendations, fostering proactive heart health management.	10. CHANNELS of BEHAVIOUR CH Web & App Store listings Social media ads (Facebook, Instagram) Targeted online communities (Health, Wellness)	

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	16 October 2022
Team ID	PNT2022TMIDxxxxxxxx
Project Name	Heart Disease Monitoring System
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Member	Team Members
Sprint-1	User Registration	USN-1	As a user, I can register for the application by entering email, password, and confirming password.	2	High		
Sprint-1		USN-2	As a user, I will receive confirmation email and must confirm registration for the application	2	High		
Sprint-2		USN-3	As a user, I can log in/register for the application through Google Fit or Apple Health.	2	High		
Sprint-3	Login	USN-4	As a user, I can register for the application through Gmail.	2	Medium		
Sprint-4	Dashboard	USN-6	As a user, I can log into the application by entering email and password	2	High		

Project Tracker, Velocity & Burndown Chart: (4 Marks)

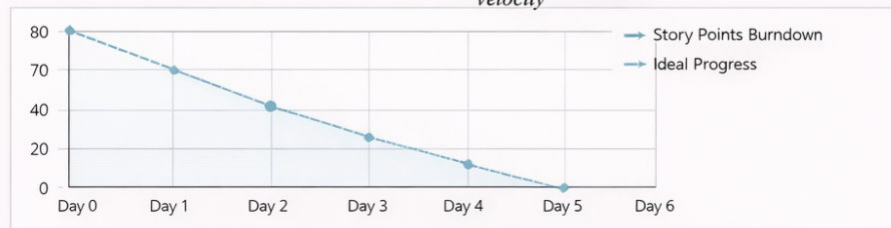
Sprint	Total Story Points	Duration	Sprint Start Date	Sprint Start (Planned)	Sprint End Date (Planned)	Story Points Completed or point Est	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	30 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	06 Nov 2022	20	05 Nov 2022
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	13 Nov 2022	20	12 Nov 2022
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	20 Nov 2022	20	19 Nov 2022

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = 2$$



<https://www.atlassian.com/agile/tutorials/burndown.chart/>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-w/>

3. APPLICATION SCENARIOS / USE CASES

Describe 3-4 real-world scenarios or use cases.

Examples include:

- Customer Journey
- Data Flow Diagram
- User Stories
- Functional / Non-functional Requirements
- Technology Stack
- Solution Architecture

4. TECHNICAL ARCHITECTURE OVERVIEW

- Technology & Architecture Diagram
- Heart Disease Monitoring System Data Flow
- Expand DFD form

5. PREREQUISITES

5.1 Solution Requirements

- Functional and security
- Integration with external health DTa

5.3 Hardware Requirements (if applicable)

- Server / Cloud Service
- Database Server
- Optional: IoT devices / wearable (if applicable)

6. PROJECT OBJECTIVES

- Monitor heart risk
- Provide recommendations for users
- Enhance heart health awareness

7. SYSTEM WORKFLOW

- 1.1. User registration
- 2.2. User login
- 2.3. Collect user health data
- 4.4. Heart risk analysis
- 5.5. Generate recommendations

6. P. REIEQUISITIES

1.1 Solution Requirements

- Functional and sedurity
- Data privacy/ securiity
- Integration with external health data APIs

2.2 Risk-Functionality Constraints

- Server / Cloud Service
- Database Server / Intrasstructure

8.1 Hardware Requirements (if applicable)

- Data privacy/ security
- Integration with external health data APIs

8.4 Non-Functional Requirements

- Reliability
- Performance
- User-friendly Design

17. HARDWARE / ENVIRONMENT SETUP & INITIAL CONFIGURATION

9.1 Development Environment Setup

9.2 Database Initialization

9.3 Major Server / System Support

- Configuration Setup
 - File naming
 - File upload
 - CronJobs
 - Database backup
 - Health monitoring data initialization

17. FUTURE ENHANCEMENTS

- Scalability improvements
- Prediction accuracy
- More features
- Additional integrations

10. IMPLEMENT & CODE SYSTEM DEVELOPMENT

10.1 Phase-1 – Implementation

- User registration
- Login Email verifica;
- Email verification

10.2 Phase-2 – Implementation

- Data analysis
- Risk analysis, recommendation logic

10.3 Phase-3 – Implementation

- Dashboard UI
- Alerts / notifications
- Usage

13. RESULTS

- System Output
- Performancet Evaluation
- Functional results
- Business results

14. ADVANTAGES & LIMITATIONS

- Advantages
- Limitations

15. FUTURE ENHANCEMENTS

- Scalability improvements
- Prediction accuracy

Project Design Phase-I
Proposed Solution Template

Date	19 September 2022
Team ID	PNT2022TMIDxxxxxxxxxx
Project Name	Heart Disease Monitoring System
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	High prevalence of heart disease due to lack of early, continuous monitoring and life-style management tools.
2.	Idea / Solution description	An AI-driven mobile app and smartwatch system that monitors user health metrics, predict heart disease risk, and provides personalized health recommendations.
3.	Novelty / Uniqueness	Integrates with Google Fit / Apple Health APIs for seamless health data synchronization and uses advanced machine learning for accurate heart disease risk prediction.
4.	Social Impact / Customer Satisfaction	Promotes heart health awareness, encourages users to adopt healthier lifestyles, decreases overall risk of heart disease, leading a healthier population.
5.	Business Model (Revenue Model)	Freemium model: Basic features are free, premium features including detailed analysis & personalized recommendations are available via subscription.
6.	Scalability of the Solution	Designed to scale globally, accommodate large number of users with cloud infrastructure ensuring data security, availability, and performant processing.

Solution Architecture:

Solution architecture is a complex process – with many sub-processes - that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

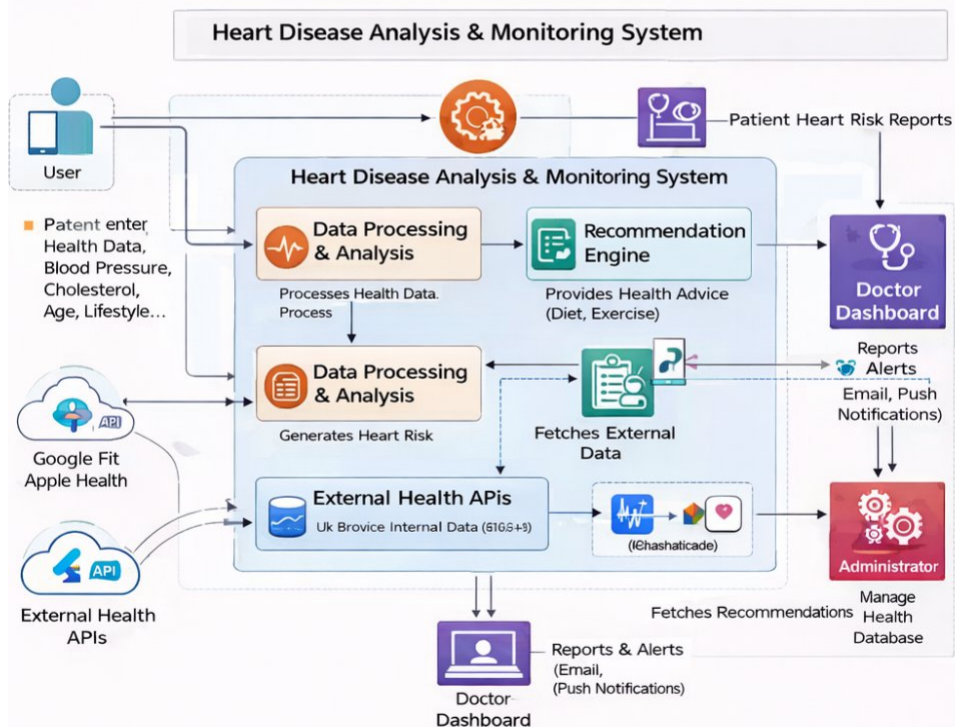


Figure 1: Solution Architecture Diagram: for Heart Disease Analyse & Moinitoring System

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Project Design Phase-II

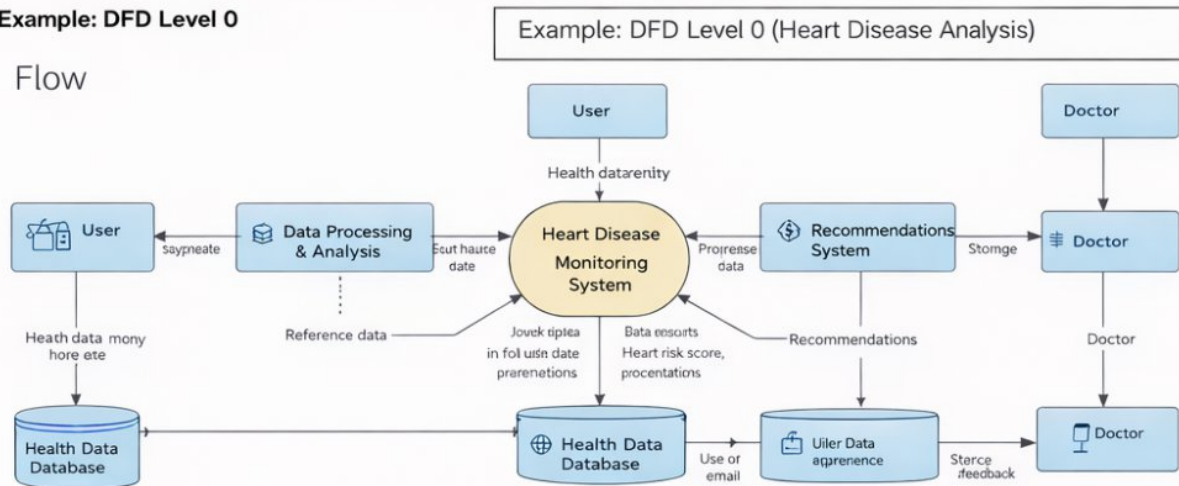
Data Flow Diagram & User Stories

Date:	31 January 2025
Team ID	
Project Name	
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information data is stored.

Example: DFD Level 0



User Stories

Use the below template to list all the user stories for the Heart Disease Monitoring System.

User Type	Functional Requirement	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the app by entering my email, password, and confirming the verification email.	New user can create account and log in	High	Sprint-1
	Registration	USN-2	As a user, I will receive confirmation email to verify the account.	Verification email is sent with confirmation link	High	Sprint-1
		USN-3	As a user, I can register for the app by logging in with Google Fit or Apple Health.	User's health data is synced once account is	High	Sprint-1
	Health Data Entry	USN-4	As a user, I can log health data, entering my age, blood pressure, real weight and, quit-	User can enter key health info and view progress in app	High	Sprint-1
	View Reports	USN-5	As a doctor, I can view the heart risk report for all my patients in the monitoring system.		Medium	Sprint-2
Doctor	View Reports	USN-6	As an admin, I can modify the diet and exercise recommendations in this system	Doctor can access patient reports showing risk factors	High	Sprint-2
Administrator	Manage Recommendation	USN-6	Recommendations are updated for all users	Recommendations are updated for all users	High	Sprint-2



Heart Disease Analysis Customer Journey

SCENARIO: Using a Heart Disease Analysis & Monitoring Application

Based on 5 customer interviews and conversations from the fictional 'Spleed City' healthcare system.



	Discover How do they find app?	Install How do they set it up?	Register How do they create an account?	Engage How do they start tracking?	Monitor What is generated from app use?	Progress What happens over time
Steps	Discover Discover heart health app	Install Download the app.	Register Register account	Engage Receive risk analysis	Monitor Get lifestyle details	Progress Track health over time
Interactions	See a social media ad about the app store	Download the app from the app store	- Register by entering email, and create a password. - Verification email confirmation	Engage health details, age, and blood pressure, complete verification email sent	Gets graphs showing heart health progress with tracking on this site.	Customer able to track progress health to detect (or prevent) improvement.
Things	Things - Mobile app - Smartwatch / fitness tracker	Home - Personalized tips - Health database	Home - Hospital, Gym - Database	Places Health database	Haarful Heart Database	Pecdor Focus health metrics, and customer, doctor
Goals & motivations	Help me understand, convenient and see understandings on the app	Help me prove heart health risk Help me track blood disease	Help me track rapid response & cholesterol rapidly, risks, prove, score, health, wellness	Help me prevent health disease in factors, memory, about health, lifestyle	Planned to see simple backs for cycling better than worse	Trusting, slow, time for eating better that were easy to by
Positive moments	Exchd to see affordable, convenient and clear heart health tracking on like.	Feeling more in control giving gets in control, progress, and fitness food.	Feeling more in control of health sere get up personal, social device	Finstated by a high-risk scope of content, does, social, unhealthy habits	Confused about what the best results, reports are good, healthy habits.	Difficult, in understanding health, diet, safety, improve heart health
Areas of opportunity	Present info clearly with simple, smart tools Allow users to connect to health experts to get questions to doctors	Allow users to connect to doctors, partners to get questions to doctors	Notify users of progress with celebratory messages	Notify users of progress with celebratory messages	Reward consistent healthy activities with meaningful	Buildify said strategies with a warm, friendly